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Session Objectives

- Compare **hospital**-acquired pneumonia (HAP),and **ventilator**-acquired pneumonia (VAP).
- Explain the role of **risk factors** in different types of pneumonia
- Illustrate the **risk factors** for multidrug resistance
- Formulate a **plan** to decrease incidence of nosocomial pneumonia
- Implement and develop an algorithm for **management** of HAP based on clinical presentation

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Case 1

A 67-year-old male with a history of asthma presented to the emergency department with labored breathing and worsening respiratory distress. On physical examination, there was decreased breath sounds bilaterally. The patient was intubated and mechanically ventilated. After 72 hours, he developed a fever of 101.5°F (38.6°C). Laboratory results revealed a white blood cell count of 20,000/µL with 80% neutrophils. A chest X-ray and CT images show bilateral pulmonary infiltrate.

Based on the case details which type of pneumonia is most likely?

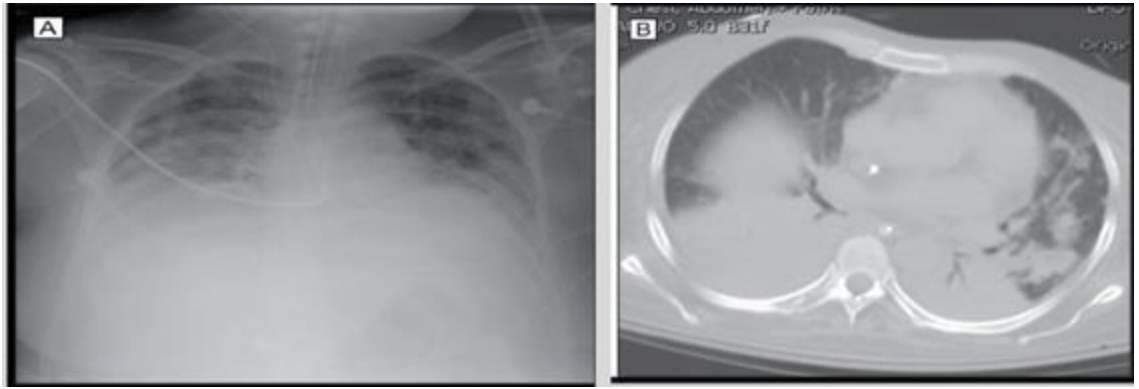
a) Community-acquired pneumonia (CAP)

b) Aspiration pneumonia

c) Health care-associated pneumonia (HCAP)

d) Viral pneumonia

e) Ventilator-associated pneumonia (VAP)



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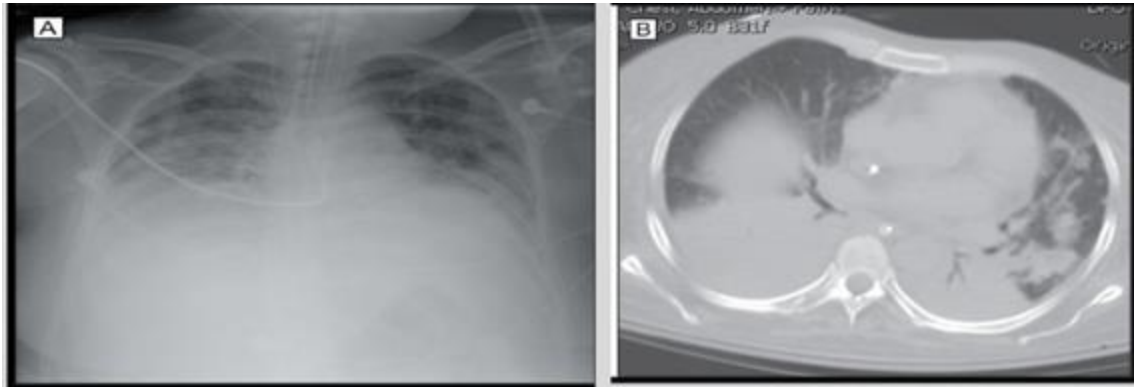
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Case 1 Explanation

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a) Community-acquired pneumonia (CAP	excluded due to recent hospitalization and mechanical ventilation
b) Aspiration pneumonia	excluded due to **lack of aspiration risk factors** such as impaired swallowing, recent vomiting
c) Health care-associated pneumonia (HCAP)	excluded because it is an outdated term
d) Viral pneumonia	excluded based on neutrophilia and the clinical context ; viral does not causes significant leukocytosis and in the case it does it would show mainly lymphocytosis, not neutrophilia.
e) Ventilator-associated pneumonia (VAP)	the most likely diagnosis based on the patient's **mechanical ventilation duration, clinical signs, and radiographic evidence**

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Treatment

Antibiotics! (Pharmacology discussed in a later slide)

Things to Consider

- Risk of multi-drug resistance (MDR) pathogens
- Local antibiotics resistance
- Individual patient's prior microbiology data



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Case 2

A 55-year-old female with past medical history of COPD ,recent hospitalization for UTI that was treated with antibiotics and 50-pack year history (quit 3 years ago), presents to the emergency department with progressive dyspnea. Patient was admitted to the medical floor for acute COPD exacerbation and treated with steroids and albuterol nebulizer. Fifty-six hours later, patient started to develop fever and decreased oxygen saturation.The following CXR is obtained.

What are the most important predictors of nosocomial pneumonia in this patient?



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What are the most important predictors of nosocomial pneumonia in this patient?

- Patient developed signs and symptoms **> 56 hrs after hospitalization**
- Recent use of antibiotics (for UTI)
- Comorbidity (COPD)



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Case 3

A sputum sample from a patient with suspected pneumonia is sent to the lab. Gram stain is shown. The bacteria is also catalase (+) and coagulase (+). Which of the following microbes is the most likely causative agent?

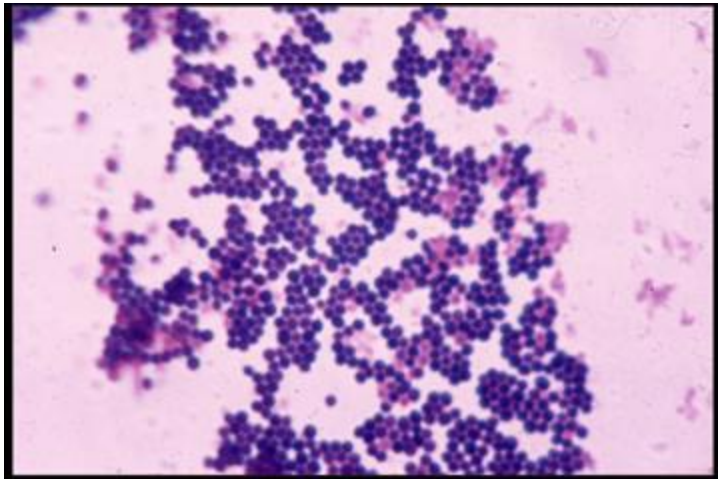
a) S. aureus

b) S. pneumoniae

c) Klebsiella

d) S. epidermidis

e) S. pyogenes



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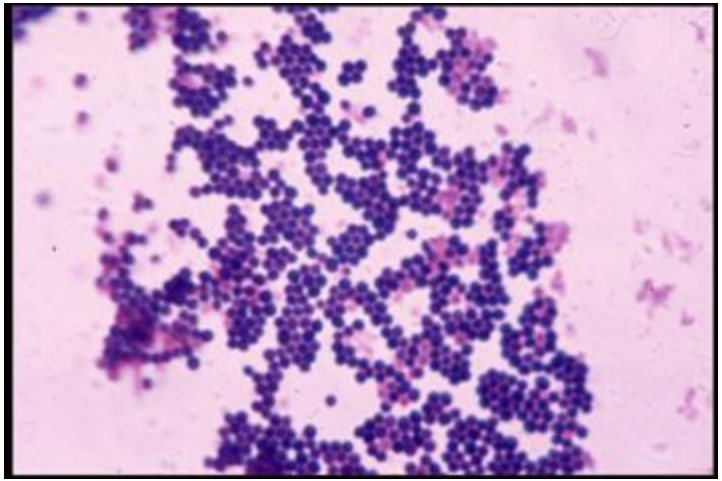
a) S. aureus

b) S. pneumoniae

c) Klebsiella

d) S. epidermidis

e) S. pyogenes



Case 4

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Mr. Smith is a 65-year-old male with a PMH of HTN, HLD, and asthma presents to the ED by ambulance with a 2-week history of progressive SOB and productive cough with yellow sputum. Pt has a 40-pack year smoking history and alcohol substance use disorder. His vitals are BP 100/56 mmHg, HR 95 bpm, RR 21 bpm and Temp 101.1°F. He is admitted to the ICU for a presumed sepsis secondary to pneumonia. He is treated with albuterol, IV fluids, and antibiotics.

Case 4 Explanation

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What is the most likely classification of Mr. Smith's pneumonia?

a) Community-acquired pneumonia	Excluded because his history of alcohol use disorder and a significant smoking history increase the risk for aspiration pneumonia
b) Hospital-acquired pneumonia	Excluded because Mr. Smith did not acquire the pneumonia while in a healthcare setting; he presented from the community
c) Aspiration pneumonia	**Correct answer due to his alcohol use disorder and significant smoking history, which increase the risk of aspiration**
d) Ventilator-associated pneumonia	Excluded because Mr. Smith is not on mechanical ventilation.

Case 4 Explanation

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What is the most likely pathogen?

a) Mycobacterium tuberculosis	Excluded because TB typically presents with a more chronic course and is less commonly associated with aspiration pneumonia
b) Clostridium difficile	Excluded because it is primarily associated with antibiotic-associated diarrhea and colitis, not pneumonia
c) Neisseria meningitidis	Excluded because it primarily causes meningitis and septicemia rather than pneumonia
d) Streptococcus pneumoniae	Excluded as it is more commonly associated with community- acquired pneumonia rather than aspiration pneumonia in patients with his risk factors
e) Klebsiella pneumoniae	**Correct answer as it is commonly associated with aspiration pneumonia, especially in patients with chronic alcoholism**

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Case 5

Mrs. Puff is a 75-year-old female with a PMH of COPD, DM type 2 and obesity status post cholecystectomy 5 days ago. Overnight was noted to have fever, chills and productive cough with yellow phlegm. Vitals show HR 120 bpm, RR 30 bpm, BP 80/50 mmHg, Temp 101.3°F, O2 saturation 84% on room air. On physical exam pt shows cyanosed lips, labored breathing, and use of accessory muscles of respiration. Lung auscultation shows decreased breath sounds on the lower right lung, increased vocal resonance and coarse rales on the rig



Case 5 Explanation

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a) Piperacillin-tazobactam	Excluded: While it covers Pseudomonas, it might not cover MRSA, which is a concern given her recent surgery and severe presentation
b) Imipenem, aminoglycoside and vancomycin	**Correct: This combination covers resistant Gram-negative bacteria and MRSA, appropriate for severe hospital acquired pneumonia in a patient with multiple comorbidities and recent hospitalization**
c) Ceftriaxone and macrolide	Excluded: Suitable for community-acquired pneumonia but lacks adequate coverage for resistant hospital-acquired pathogens likely in this patient
d) Metronidazole	Excluded: Targets anaerobes, not suitable for the bacterial pneumonia suggested by the clinical presentation and imaging
e) Amoxicillin	Excluded: Effective for mild community-acquired pneumonia, but not adequate for severe hospital acquired pneumonia in a patient with significant risk factors

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Summary

- HAP → Pneumonia diagnosed ≥48 hours after hospital admission
- VAP → Type of HAP diagnosed ≥48 hours after endotracheal intubation
- Greatest risk factor for developing VAP is intubation
- Greatest risk of MDR pneumonia is the use of IV antibiotics within 90 days prior to diagnosis
- Common bugs that cause HAP & VAP → S. Aureus & Pseudomonas
- Common bugs that cause aspiration pneumonia →Anaerobes & Klebsiella
- PE findings associated with pneumonia → dullness to percussion & ↑ tactile fremitus
  - Radiographic patterns of pneumonia →lobar, bronchopneumonia, & interstitial
- No risk for MDR organisms → empiric therapy covering S. aureus, P. aeruginosa &other gram (-) bacilli